**Course Project:** Bookstore Data Warehouse & Business Intelligence

**Objective:** Design, implement, and analyze a Star Schema for a bookstore to support data-driven decision-making.

## 1. Project Overview

You are appointed as a Data Warehouse Consultant for a growing bookstore. The management wants to move from simple record-keeping to **Strategic Analysis**. Your task is to transform their operational data into a structured Data Warehouse (Star Schema) and extract actionable insights.

## 2. Phase I: Dimensional Modeling

**Task:** Based on the provided entities, design a Star Schema.

- **Fact Table:** Fact_Orders (contains measures like quantity/price).

- **Dimension Tables:** Dim_Book, Dim_Client.

- **Requirement:** Add a Date Dimension (Dim_Date) to the schema to allow time-based analysis (Years, Quarters, Months).

## 3. Phase II: Schema Implementation (DDL)

Write SQL scripts to create the environment:

- Define **Primary Keys** for all dimensions.

- Define **Foreign Keys** in the Fact table to ensure referential integrity.

• **Bonus:** Implement a **Slowly Changing** Dimension (**SCD Type 1**) logic for the Dim_Book table when updating prices.

## 4. Phase III: Data Ingestion (ETL)

Populate your Warehouse with the following initial dataset:

• **Books:** (301, 'Python', 'Java', 80), (302, 'Data Intro', 'Data', 120).

• **Clients:** (20, 'Ahmad', 'G'), (21, 'Sara', 'R').

• **Orders:** (1, 301, 20, 80), (2, 302, 21, 120).

## 5. Phase IV: Analytical Reporting (SQL)

Answer the following Business Questions using SQL queries:

• **Inventory Performance:** List the top 2 books in 'Java' or 'Data' categories with a price > 50, sorted by the highest price.

• **Sales KPIs:** Calculate the following metrics from the Fact_Orders:

• Total Revenue (Total Sales).

• Average Order Value (AOV).

• Total Number of Transactions.

• Min/Max Transaction Value.

• **Product Contribution:** Identify books that have generated a total revenue exceeding 90 USD.

## 6. Project Deliverables

• **ER Diagram:** A visual representation of the Star Schema.

• **SQL Script File:** Containing all CREATE, INSERT, UPDATE, and SELECT statements.

• **Final Report:** A brief explanation of how this Data Warehouse helps the bookstore owner increase profits.

-*The project was submitted as a PDF file containing screenshots of the code results.*